

Weiqi (Rocky) Wang

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EDUCATION

University of California, Los Angeles, Ph.D., Mechanical Engineering (Robotics)

September 2026

Dissertation: Embodiment-Aware Task Realization for Human-Robot Coexistence

Johns Hopkins University, M.S., Mechanical Engineering (Robotics)

May 2017

University of Miami, B.S., Mechanical Engineering; Minor in Mathematics

May 2015

SKILLS

- **Programming Languages:** Python; C++; MATLAB.
- **Software Tools:** PyTorch; ROS/ROS 2; CasADi; Linux; Git; SolidWorks.
- **Robot Learning:** Imitation learning and behavior cloning; reinforcement learning; sim-to-real deployment.
- **Language Models:** LLM/VLM integration; vision-language-action (VLA) training.
- **Planning:** Motion planning; trajectory optimization; constrained nonlinear optimization.
- **Control:** Robot kinematics and dynamics; system identification; model predictive control (MPC); optimal control; impedance control; admittance control; teleoperation.
- **Perception:** 6D pose estimation; state estimation; sensor fusion; point-cloud registration; 3D reconstruction; camera calibration.
- **Machine Learning:** Graph neural networks; probabilistic modeling.
- **Simulation:** Isaac Lab; Isaac Sim; MuJoCo; PyBullet; Gazebo; Webots; CoppeliaSim.
- **Hardware Interfaces:** CAN bus; Universal Robots UR5; Franka FR3; Kinova Gen3.
- **Languages:** Fluent in English, Mandarin Chinese, and Cantonese.

WORK EXPERIENCE

OffWorld, Inc. | Pasadena, California | *Software Intern*

June – September 2023; June – September 2021

- Developed a **computational geometry simulation framework** in **Python** for prototype autonomous underground mining robots, modeling environmental changes caused by robot sawing actions.
- Built **ROS/ROS 2 simulation testbeds** using Gazebo and RViz for multi-robot coordination and navigation.
- Procedurally generated **Webots scenarios** to evaluate robot performance across diverse conditions.
- Modeled tool-mounted serial-manipulator dynamics and enabled task-specific **force-feedback control**.
- Developed a **dual-arm motion-planning framework** using sampling-based and optimization-based methods, achieving **100% planning success** in the evaluated scenarios.

CloudMinds/INNFOS Technology Ltd. | Beijing, China | *Algorithm Engineer*

December 2017 – September 2019

- Led the development and maintenance of **ROS control software** in **C++** for the XR1 dual-arm mobile humanoid robot.
- Designed and implemented wearable-IMU motion capture for **real-time teleoperation** of a **34-degree-of-freedom humanoid robot**, enabling manipulation tasks such as pouring coffee.
- Developed and optimized **dynamic parameter identification** for dual-arm robots on mobile platforms.
- Applied the identified dynamics model to implement dynamic compensation, **collision detection, impedance control, and admittance control** for physical human-robot interaction.
- Tuned **position, velocity, and current control loops** for robot joint actuators, adjusting servo gains to reduce tracking lag, overshoot, and oscillation and improve response under varying loads.
- Engineered a **quadruped locomotion control** system in C++ that generates walking and trotting gaits and controls body position and orientation in six degrees of freedom.
- Built quadruped simulation testbeds and optimized design parameters based on simulated performance analysis.

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RESEARCH EXPERIENCE

Center for Vision, Cognition, Learning, and Autonomy | UCLA | *Graduate Student Researcher* Sep 2020 – Present

SUN: Agentic Robot Policy Learning with Persistent Task Programs

- Built Kuafu, an **agentic robot-learning harness** that turns object videos and language instructions into learned manipulation policies; designed shared objectives linking optimal control and reinforcement learning.
- Developed a closed-loop **agent-driven workflow** for task decomposition, program verification, diagnostics, and repair, achieving a 95.6% formation success rate.
- Achieved **82.03% mean success** on nine simulated multi-stage tasks via behavior cloning and **residual RL**; outperformed evaluated LLM reward-design and zero-shot language-guided planning methods by 69.18 and 72.03 percentage points.
- Built a web-based **teleoperation application** to collect human demonstrations in simulation, using AgileX PIKA for task-space robot control.
- Built a demonstration collection pipeline using learned controllers, achieving **10.57×** human teleoperation throughput, breaking even on cost at 35 and time at 513 demonstrations per task on average (including controller preparation).
- Trained RGB, point-cloud, and **multi-task VLA policies** (SmolVLA, WALL-X, $\pi_{0.5}$) on Kuafu-generated demos; 3D Diffusion Policy (DP3) reached 46.02% mean success on nine tasks, 23.6 percentage points above baseline.
- Developed the DP3 deployment pipeline, achieving **zero-shot sim-to-real transfer** on Franka FR3 and Kinova Gen3 across three robot–gripper configurations without real-world demonstrations or fine-tuning.

Scene Rearrangement for Human–Robot Co-Activity

- Led development of a **scene-rearrangement system** to improve indoor navigation and object access for robots while preserving furniture arrangements that support everyday human activities.
- Implemented **hierarchical optimization** (ASA/CMA-ES) to reposition furniture, using robot footprint and **reachability models** alongside learned spatial relationships and ConceptNet semantics.
- Improved robot accessibility by **14%**, with **30%** more reachable objects in evaluation.

Human Activity Recognition and Prediction

- Designed and trained **graph neural networks** for human activity recognition and prediction in shared indoor environments.

Collaborative Motion Planning and Tool Use

- Contributed to **optimization-based motion planning** for mobile manipulation using Virtual Kinematic Chains, which combine the mobile base, arm, and manipulated object into one kinematic model to coordinate their motion.
- Developed **trajectory optimization** for tool-use motions on a simulated Baxter robot and verified the formulation in **MATLAB**; explored how different robot kinematics affect feasible tool-use strategies.

TEACHING EXPERIENCE

Mathematics for Life Scientists | UCLA | *Instructor & Teaching Fellow*

Sept 2020 – Sept 2025

- Led teaching and grading logistics for 6 years, developed curriculum, and applied evidence-based instruction; nominated for UCLA's university-wide **Distinguished Teaching Award for Teaching Assistants**.

EARLIER RESEARCH EXPERIENCE

Laboratory for Computational Sensing + Robotics | Johns Hopkins University | *Research Assistant* Jun 2016 – Sep 2017

Fiber Bragg Grating Sensors

- Evaluated **fiber Bragg grating shape sensors** for dexterous surgical continuum manipulators.
- Developed a **MATLAB graphical interface** for real-time 3D reconstruction of needle kinematics and integrated low-level control into the experimental setup.

Collaborative Robotic System for Ultrasound Imaging

- Investigated collaborative robotic methods to reduce the physical pressure required during ultrasound imaging.
- Contributed to **optimal-control-based virtual fixtures** that constrain robot motion to support positioning accuracy and procedural safety during collaborative ultrasound imaging.

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Haptic Simulation for Femur Bone Drilling

- Led a team developing a **virtual femur-drilling training environment** for medical residents.
- Built the simulator in **CHAI3D and Visual Studio**, integrating 3D anatomical models with interactive haptic simulation.
- Automated **femur CT segmentation** and constructed anatomically accurate mesh and voxel models for the simulation.
- Implemented **haptic feedback** that allowed users to distinguish anatomical structures while drilling the virtual femur.

da Vinci Research Kit: Master-Arm Gravity Compensation

- Implemented **model-based gravity compensation** for the **da Vinci Research Kit master arm** using analytically identified inertial parameters.

SELECTED PUBLICATIONS

- Wang, W., et al. “SUN: Agentic Robot Policy Learning with Persistent Task Programs.” *Under Review*, 2026.
- Wang, W., et al. “Rearrange Indoor Scenes for Human–Robot Co-Activity.” *IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 11943–11949.
- Zhang, Z., et al. “Understanding Physical Effects for Effective Tool-Use.” *IEEE Robotics and Automation Letters* 7(4), 2022, pp. 9469–9476.
- Jiao, Z., et al. “Efficient Task Planning for Mobile Manipulation: A Virtual Kinematic Chain Perspective.” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021, pp. 8288–8294.
- Zahedi, E., et al. “Towards Skill Transfer via Learning-Based Guidance in Human–Robot Interaction: An Application to Orthopaedic Surgical Drilling Skill.” *Journal of Intelligent & Robotic Systems* 98, 2020, pp. 667–678.